Image analysis (MicroICS) hands-on session: Software installation instructions

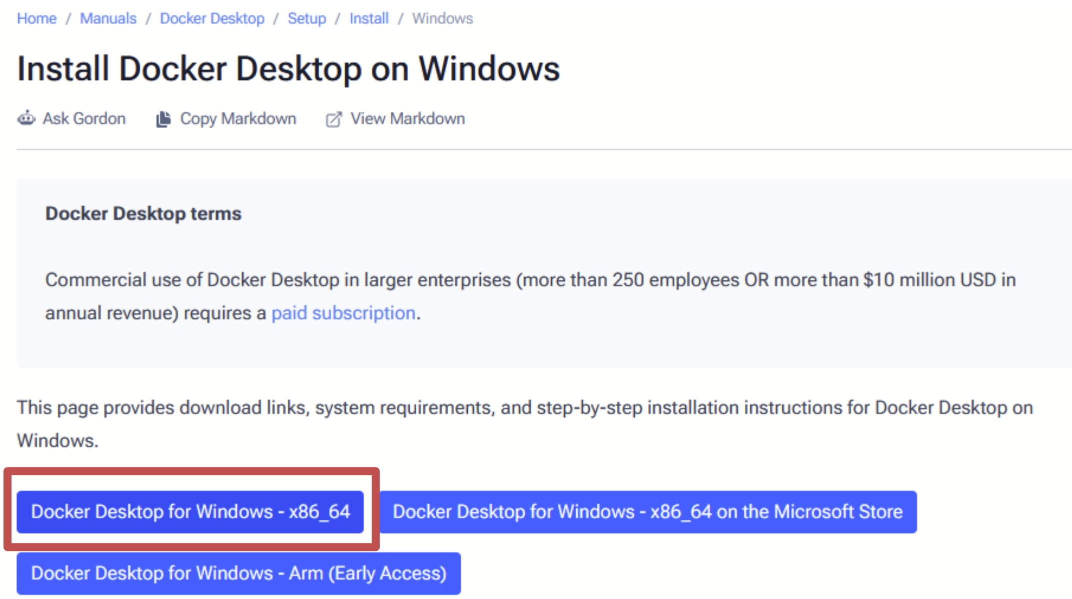
Part of: "Deep dive into the statistical approaches used to validate potential standardised laboratory methods"

In this hands-on session, we will learn how to use MicroICS to analyse biofilm images. It runs as a self-contained Docker image (https://hub.docker.com/r/skblaz/microics) accessed through a browser-based interface, and the source code is available on GitHub (https://github.com/SkBlaz/biofilm-classification).

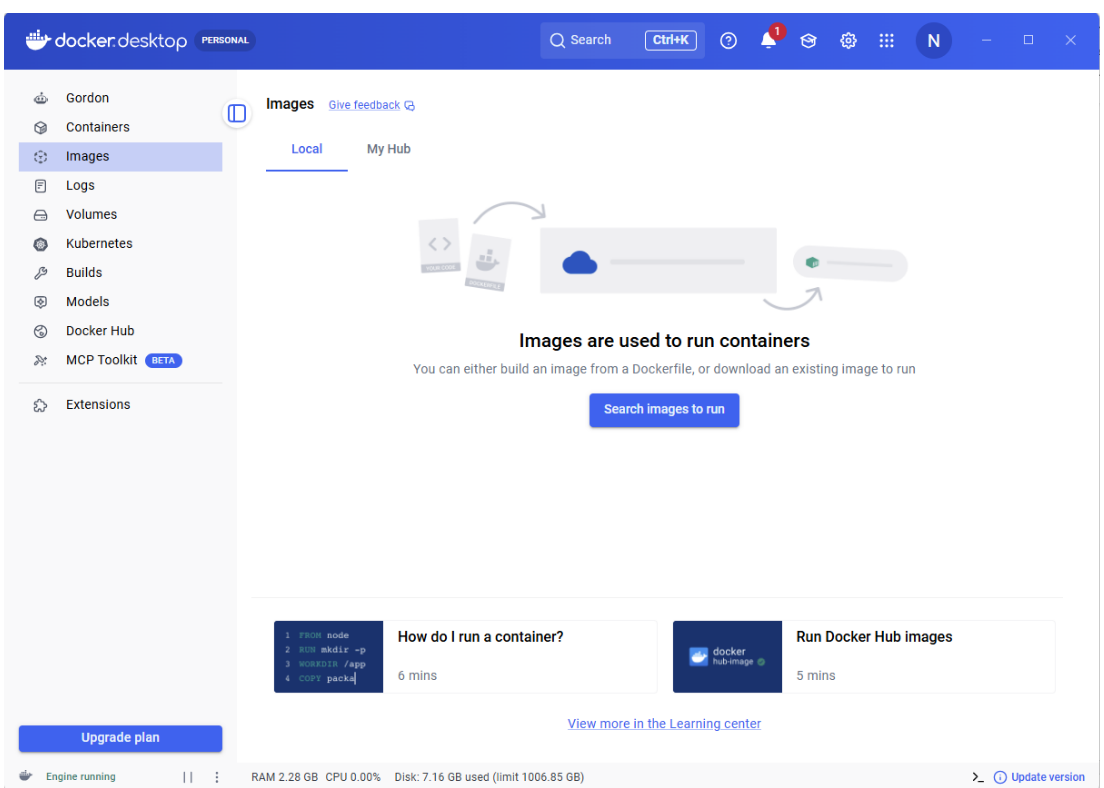
Please install Docker before the workshop and launch it once to confirm that it starts without warnings. On some institutional laptops, Docker requires system settings that only your IT department can enable, so checking this in advance will avoid delays on the day.

Installation on Windows

- **Install Docker Desktop**
  - On the Docker download page (https://docs.docker.com/desktop/setup/install/windows-install/), click the button for your version of Windows (amd64 or arm64). The download will start.



  - When it finishes, double-click the downloaded file to begin installation.
  - Follow the installer, using the default (recommended) options.
  - Once installed, start Docker Desktop and wait until the whale icon indicates that it is running. Active Docker Desktop looks like this:

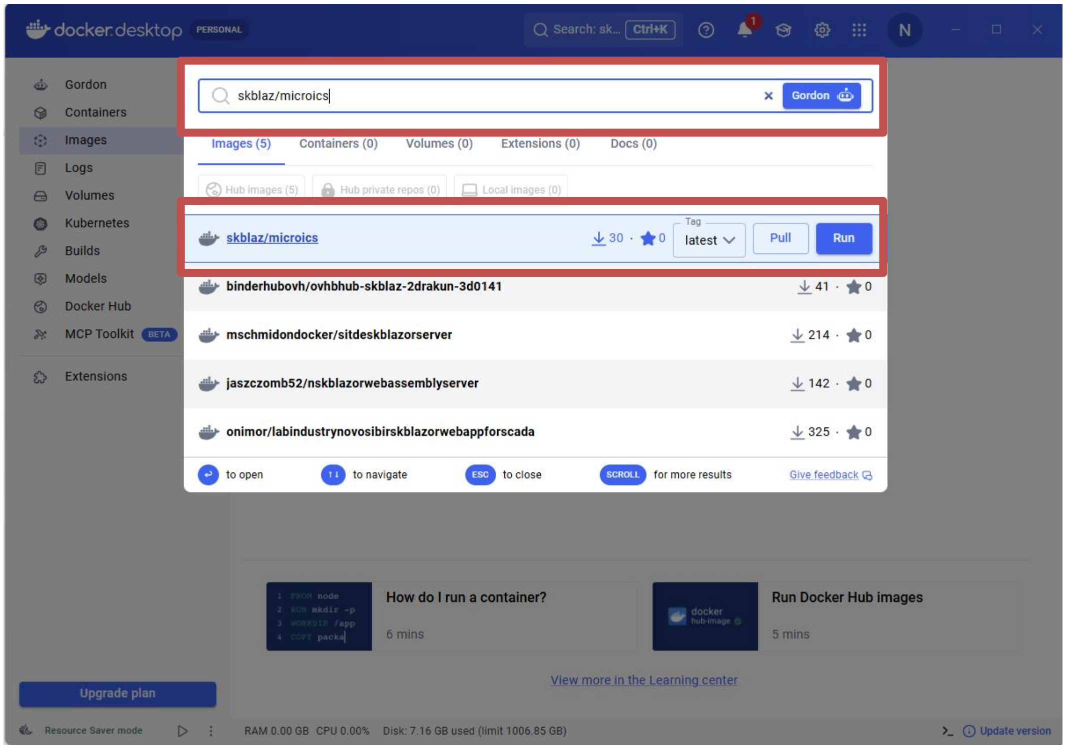


● Setting up MicroICS (via Docker)

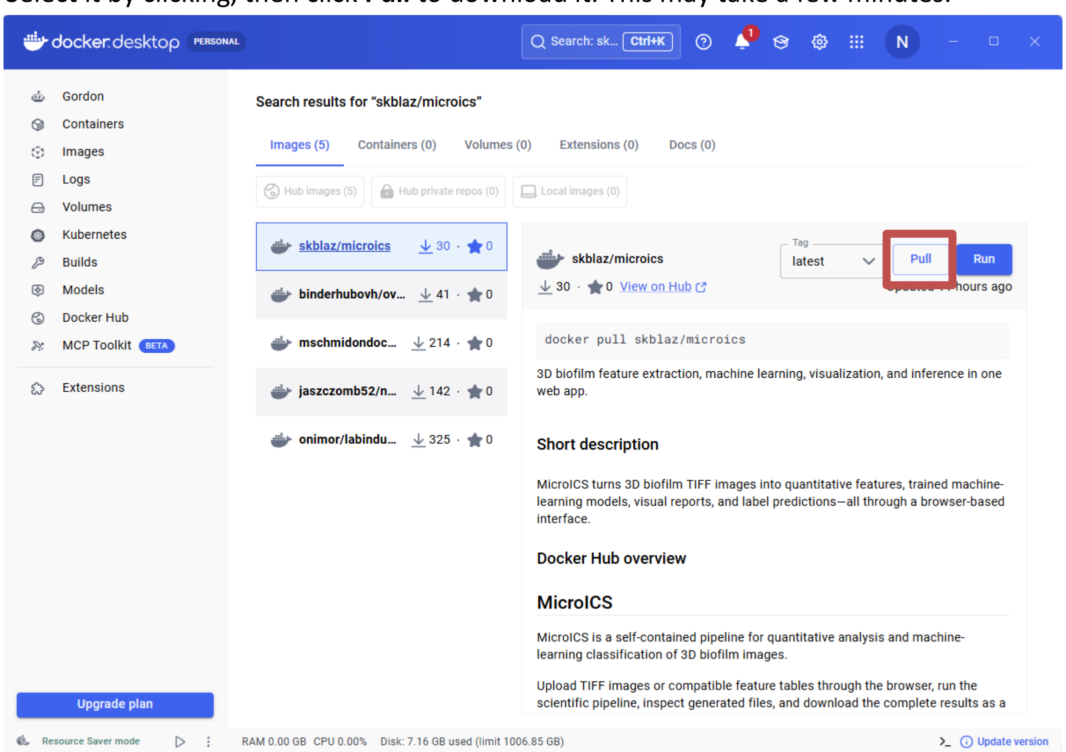
You can do this in two ways: using Docker Desktop (the graphical interface – make sure it is running and start it if not), or using the command line (PowerShell on Windows, Terminal on macOS/Linux).

Option 1 — using Docker Desktop (graphical)

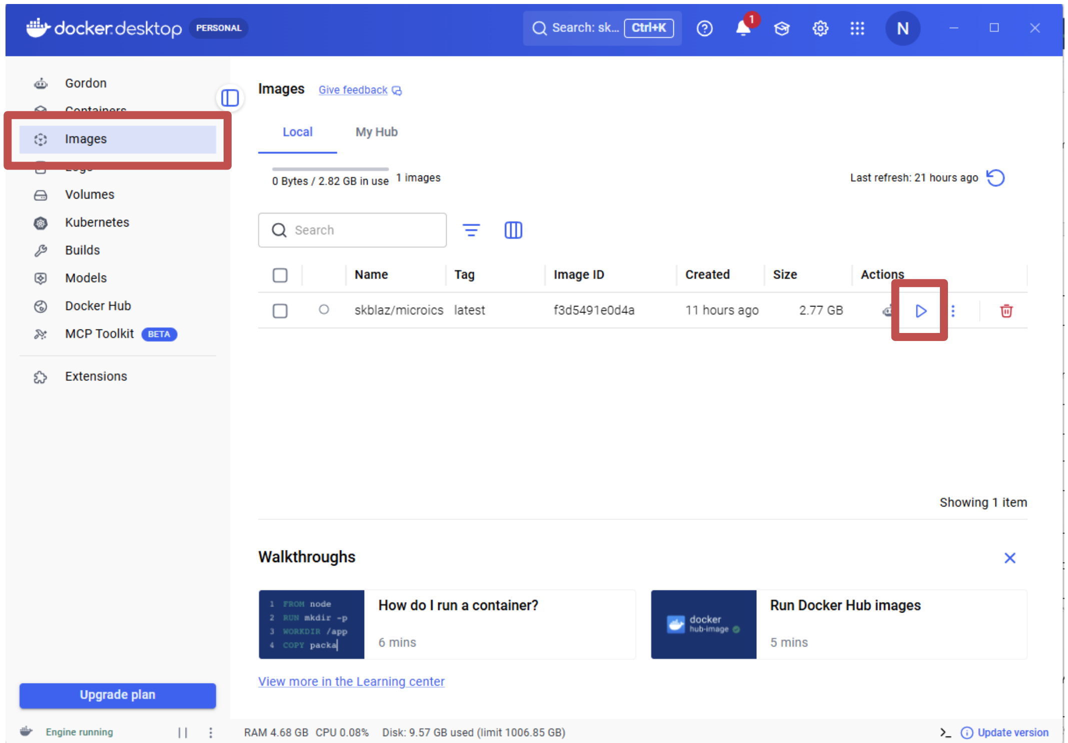
- In the search bar at the top of Docker Desktop, type **skblaz/microics**.



- Select it by clicking, then click **Pull** to download it. This may take a few minutes.

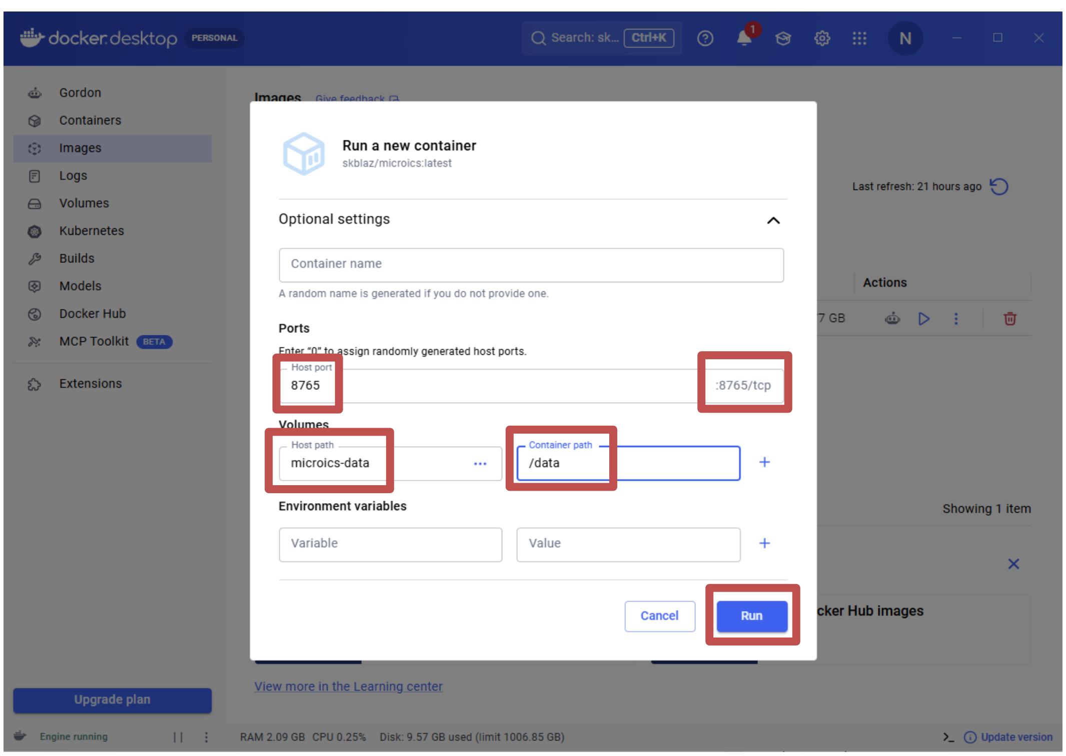


- When it has downloaded, open the **Images** tab on the left. Find skblaz/microics and click **Run** (the ▶ arrow on the right).

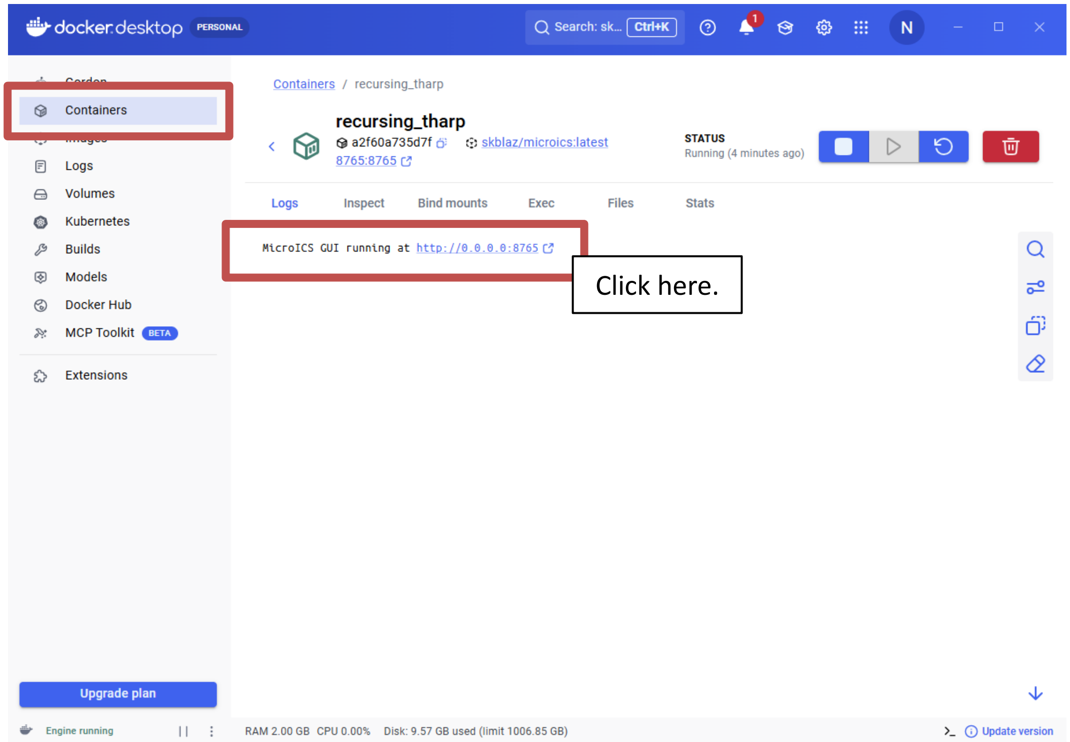


- In the dialogue that appears, click **Optional settings** to expand it, and enter:
  - **Ports** — Host port: 8765 (Container port: 8765, in grey, usually automatically filled)
  - **Host path**: microics-data, **Container path**: /data

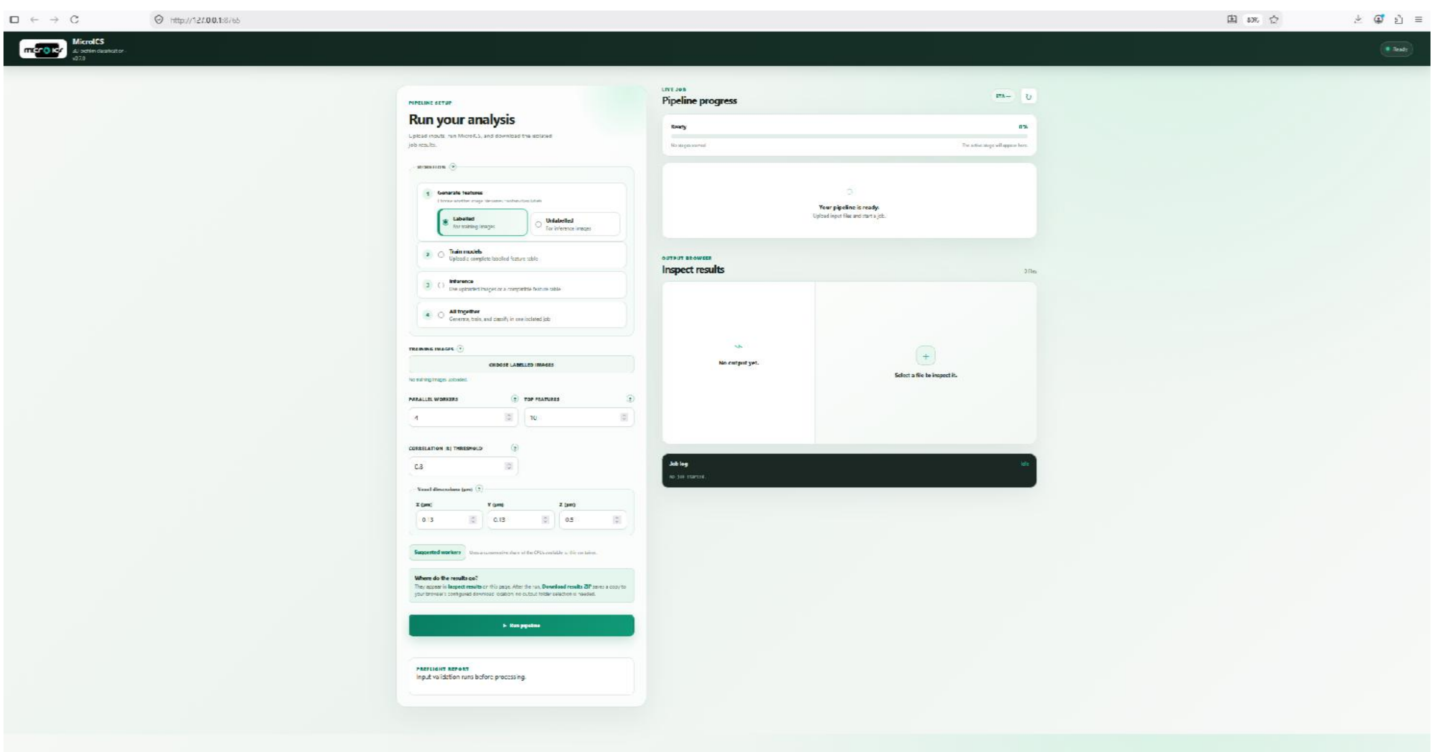
After you enter these, click **Run**.



- The container now appears under the **Containers** tab. Docker Desktop shows a clickable **8765:8765** link next to the running container that takes so to the MicroICS interface.



The MicroICS interface looks like this:



Option 2 — using the command line

- Make sure Docker Desktop is running, then open PowerShell (Windows) or Terminal (Mac/Linux).
- Download the image:

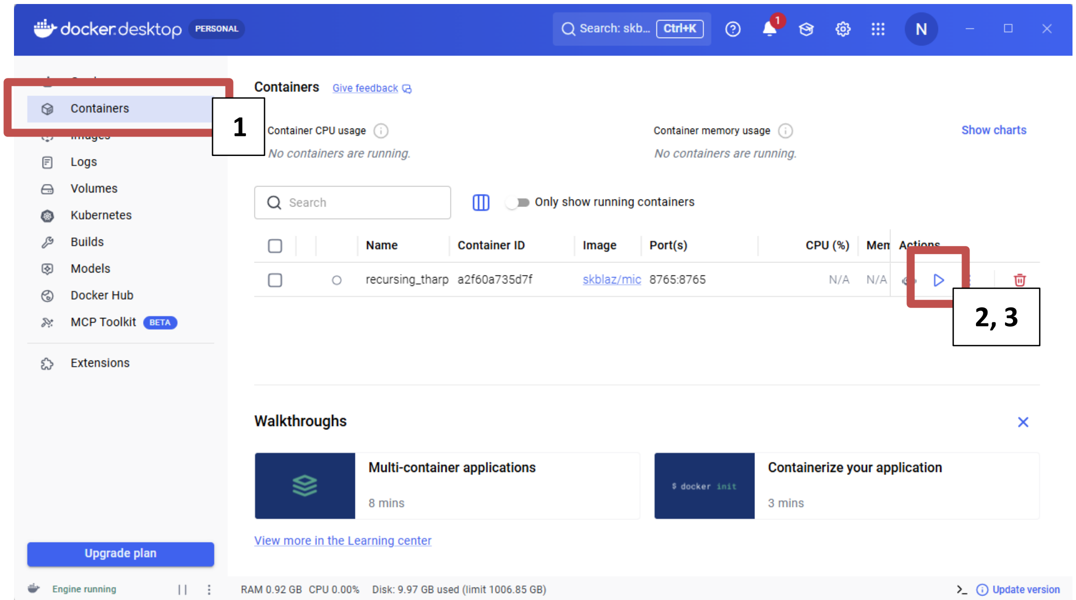
```
docker pull skblaz/microics:latest
```

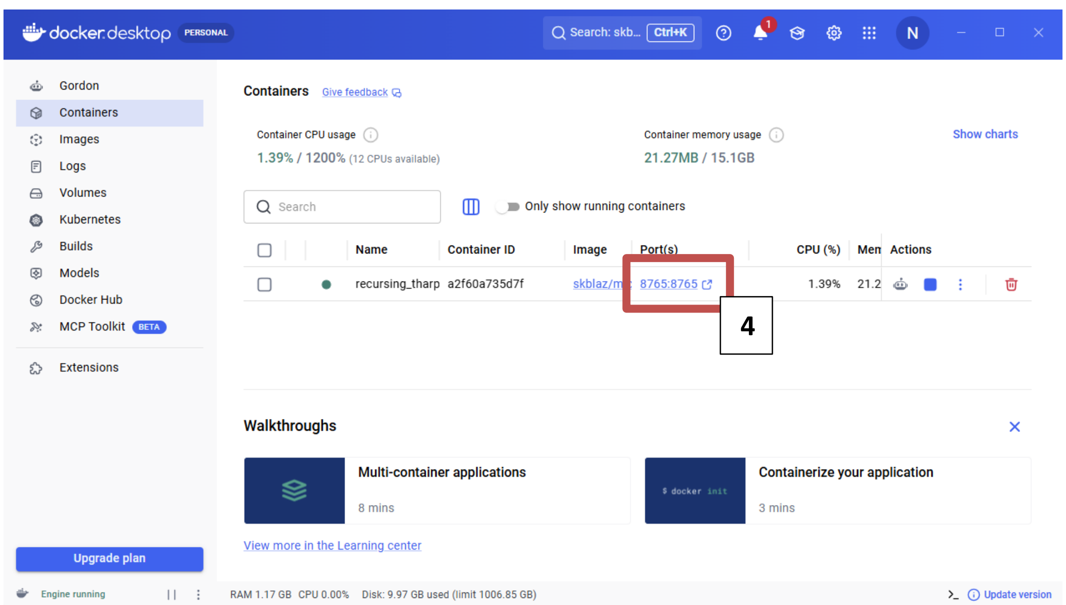
- Start the tool:

```
docker run -p 8765:8765 -v microics-data:/data skblaz/microics:latest
```

- Open http://localhost:8765 in your browser.

- **Restarting the existing container of MicroICS**
    1. Start Docker Desktop and go to Containers.
    2. Find your MicroICS container (it'll be stopped).
    3. Click the Start button (▶) next to it.
    4. Once it's running, click the 8765:8765 link, or open http://localhost:8765 in your browser.





Installation on Linux

- **Install Docker**
  - Download and run Docker's official installation script:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

```
sudo sh get-docker.sh
```

  - Allow Docker to run without sudo:

By default, Docker requires sudo for every command, which causes permission problems with output files. Add your user to the Docker group so Docker can be run directly:

```
sudo usermod -aG docker $USER
```

  - Then reboot for this change to take effect:

```
sudo reboot
```

  - Check that Docker installed correctly (recommended)

```
docker --version
```

It should print a version number.

Then continue to the "Installing MicroICS with Docker" steps — from that point on, Linux is identical to the Windows (pull the image, run it, open the browser).

Installation on macOS

- **Install Docker**
  - Download Docker Desktop from https://www.docker.com/products/docker-desktop and install it. On Mac you may be offered an Apple chip version (M1–M4) or an Intel chip version – check which you have via → About This Mac → look at "Chip" or "Processor". Install using the default options.
  - Start Docker Desktop: Open Docker Desktop from Applications and wait until the whale icon in the menu bar shows it is running. Docker must be running whenever you use MicroICS.
  - Check Docker is working (optional): Open Terminal and run:

```
docker --version
```

It should print a version number.

Then continue to the "Installing MicroICS with Docker" steps — from that point on, Mac is identical to the Windows or Linux (pull the image, run it, open the browser).

Viewing 3D images with Fiji/ImageJ – optional

To view the biofilm image stacks, you can use Fiji – a free image-analysis programme based on ImageJ, bundled with the plugins needed to open microscopy files and display them in 3D. Fiji includes its own copy of Java, so nothing else needs to be installed alongside it.

Windows

- Download the Windows version from https://fiji.sc/.
- It arrives as a .zip file. Right-click it and choose Extract All – do not run it from inside the .zip.
- Move the extracted Fiji.app folder to a permanent location, such as your Desktop or Documents (not the Downloads folder).
- Open the folder and double-click ImageJ-win64.exe to start Fiji.
  If Windows warns that the publisher is unknown, choose More info → Run anyway – Fiji is safe; the warning appears because it is not a signed commercial installer.

Linux

- Download the Linux version from https://fiji.sc/.
- Extract the .zip to a permanent location, for example your home folder: unzip Fiji.zip -d ~/
- Start Fiji from the extracted folder: ~/Fiji.app/ImageJ-linux64

macOS

- Download the macOS version from https://fiji.sc/.
- Open the downloaded .zip if it does not unzip automatically, and drag Fiji.app into your Applications folder.
- The first time you open it, macOS may block it as being from an unidentified developer. If so, right-click (or Control-click) Fiji.app → Open, then confirm. After the first time, it opens normally.